

BUILD RAIL LOCALLY - RETAIN SKILLS - REDUCE FOREIGN-CAPITAL DEPENDENCE

Give your city a railway it can own.

OpenSourceRail is an open-source, deterministic urban-rail platform designed to retain design authority, software, fabrication, integration, operations and maintenance capability in-country instead of importing a closed foreign-turnkey system.



Promoted reference: light-metro-3car

49.5 m | 360 AW2 / 480 AW3 | 675 kWh gross

THE PHYSICAL RAILWAY - MODULAR TRAIN, SIMPLE STATIONS, LOW-COMPLEXITY CIVIL WORKS, LOCAL PRODUCTION

265

DEVELOPING-WORLD CITY MODELS

43

COUNTRY PROGRAMMES

75.4%

MODELED DOMESTIC VALUE



MODULAR TRAIN

3-car driverless light-metro reference.
Replaceable body modules, battery traction and roof-PV MPPT are carried on a common car platform.



SIMPLE STATIONS

Prefab solar-canopy stations, direct pedestrian-level access where possible, and a 500 kW standard charging cabinet.



AT-GRADE
($\geq 70\%$)

ELEVATED / VIADUCT
($\leq 25\%$)

LOW-COMPLEXITY CIVIL

Target civil mix: at-grade $\geq 70\%$, elevated $\leq 25\%$, bridge $\leq 5\%$. No tunnels in the upstream reference system.



LOCAL PRODUCTION

120 LM3 product rows, 26 assembly nodes and 30 tooling/mould families connect design to repeatable work packages and QA gates.

EVIDENCE BOUNDARY

Repository outputs are planning and engineering-screening evidence - not bids, construction releases, safety certificates, approvals or endorsements.

ONE CONNECTED, REPRODUCIBLE WORKFLOW

From city map to an operating railway.

The same deterministic workspace connects city design, GIS, CAD/IFC, simulation, ERPNext business execution, FUXA supervision, costs and railway assurance - tied to the same city, asset and engineering-revision context.



FIRST ADOPTABLE PRODUCT

Start with an existing depot, workshop or pilot corridor.

ERPNext + Frappe HR + railway Ops Core + simulator + asset/evidence portal can structure assets, work orders, manufacturing packages, QA hold points, maintenance schedules, defects and acceptance evidence before any train-control software is certified.

SAFETY BOUNDARY

Keep train control in simulation or shadow mode until deployment evidence is independently assessed.

DESIGN & DELIVERY

SERIOUS DEMONSTRATION

City GIS, alignment, cost, IFC, procurement, schedule and project twin.

TRAIN + INFRASTRUCTURE

ENGINEERING DEVELOPMENT

Local-manufacture planning, supplier RFQs, prototype and civil option studies.

OPEN GOA 4 CONTROL

R&D / PRE-CERTIFICATION

Simulation, shadow mode, formal review and HIL - not live railway command.



REVIEW OR COLLABORATE

Open the public repository

Review assumptions, reproduce the generators, inspect the evidence or propose an evidence-backed contribution.

github.com/modemecotech/OpenSourceRail